

Geometric Limits on Semantic Expressibility: Quantum-Inspired Contexts and Polysemy in Large Language Models

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A fixed-width language-model representation may encode a semantic feature dictionary larger than its usable carrier dimension while exposing only a context-dependent subset at any one time. Let d denote the dimension of a declared usable carrier and let N denote the number of feature directions obtained by a prespecified extraction rule. We study the overcomplete regime $N > d$ as a modelling premise, not as a capacity result. For an isotropic random dictionary, extreme-value concentration predicts

$$\mu_{\max}^{\text{null}} \approx 2\sqrt{\frac{\log N}{d}},$$

so a chosen overlap tolerance ε defines the crowding scale $N_{\text{crowd}}(d, \varepsilon) \approx \exp(d\varepsilon^2/4)$. This is a calibrated null prediction, not a hard upper bound on learned or optimized codes. Simultaneous linear readout imposes a different constraint: for k random-like active directions, interference scales as $\sqrt{k/d}$. The sharper semantic prediction is that learning allocates smaller overlaps to features that frequently coactivate, thereby reducing coactivation-weighted interference relative to isotropic and activity-matched controls.

On this classical baseline we formulate a separate, stronger model of polysemy. Context observables share a token-associated spectral projector but act differently, and potentially noncommutatively, on sense subspaces in its orthogonal complement. Conditional projection then produces a contextual embedding. The model is quantum-inspired rather than physically quantum, and its scientific content is the testable claim that the shared-projector structure improves held-out prediction, compression, or order-sensitive contextual readout relative to parameter-matched classical alternatives.

I. INTRODUCTION

This chapter examines contextual representation in large language models (LLMs) as part of the volume’s broader inquiry into “quantum thinking.” Only a minimal orientation to LLM operation is needed here. Input text is first divided into tokens, and each token is mapped to an initial vector. Transformer blocks then use self-attention and feed-forward transformations to turn those initially context-free vectors into sequence-dependent hidden states [1]. An output layer converts the final state into token scores, and a softmax distribution supports probabilistic next-token selection. This compact chain—tokens, embeddings, contextual transformation, and finite readout—supplies the computational background for the argument below; the details of pre-training, fine-tuning, reinforcement learning, and backpropagation are not required.

Current Transformer-based LLMs are classical computational systems. They run on classical hardware and use real-valued matrix operations, normalization, nonlinearities, copying, and generally irreversible updates. Their reliance on vectors, inner products, high dimension, or probabilistic output does not by itself make them quantum or even quantum-like. The more limited question pursued here is where ordinary high-dimensional geom-

etry already explains the coexistence and contextual selection of semantic features, and where a stronger comparison with quantum-like formalisms might begin. Such a comparison would have to do more than rename familiar neural operations: it would need to identify a role for contextual bases or subspaces, incompatible or non-commuting alternatives, interference, or measurement-like readout that yields explanatory or predictive value beyond a classical dynamic-embedding account.

Semantic concurrency makes this boundary concrete. A model must preserve many latent possibilities while making only a small, context-selected subset operationally accessible. Polysemy is the simplest illustration: the same token can participate in financial, river-edge, or seat/bench contexts of “bank,” yet the surrounding sequence drives it toward different contextual representations. Here polysemy concerns related senses and is distinguished from homonymy [2]. The verb in an aircraft “banks” supplies a fourth sense, mentioned below but not used in the three-arm schematic. This relation among possibility, context, and readout connects directly to the volume’s recurring themes without implying that the underlying machine is physically quantum. It also motivates a separation between what a representation can geometrically accommodate and what a finite channel can reliably recover.

The claims below occupy different rungs of that evidential ladder. Frame coherence, spherical concentration, and correlation-readout estimates are standard mathematics; here they serve only as calibrated classical nulls

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subject to stated assumptions about the usable carrier, feature extractor, metric, and coactivation. The reference-subspace organization is a stronger empirical hypothesis about LLM internals and is tested against classical controls. The comparison with quantum contextuality remains a limited structural correspondence; no physical quantumness, Born-rule dynamics, or semantic Kochen–Specker theorem is inferred.

The geometric question begins only after an overcomplete semantic dictionary has been operationally identified. Let d be the dimension of the usable carrier after a declared restriction or whitening procedure, and let N be the number of distinct feature directions returned by a prespecified dictionary-extraction and duplicate-removal rule. We study the case $N > d$. This inequality merely specifies the domain in which exact mutual orthogonality is impossible; it is not itself a finding, and no semantic deficiency is attributed to representations with $N \leq d$.

The first substantive question is quantitative: at fixed d and N , how much interference should an unstructured overcomplete code produce, and does training reorganize that interference in a semantically useful way? For an isotropic random dictionary, the largest of order N^2 pairwise overlaps is expected to scale as

$$\mu_{\max}^{\text{null}} \approx 2\sqrt{\frac{\log N}{d}}. \quad (1)$$

Equivalently, a prescribed tolerance ε defines the random-code crowding scale $N_{\text{crowd}}(d, \varepsilon) \approx \exp(d\varepsilon^2/4)$. This is a null benchmark, not a hard limit on spherical codes or learned representations.

Worst-pair overlap alone is not a semantic criterion. Two features that are never jointly required may overlap without impairing readout, whereas a much smaller overlap between frequently coactive features can be costly. If k random-like directions are active with order-one amplitudes, the baseline linear-readout interference is

$$\sigma_{\text{int}} \sim \sqrt{\frac{k}{d}}. \quad (2)$$

The sharper prediction developed below weights squared overlaps by empirical coactivation. At matched d , N , activity, and feature importance, a semantically organized dictionary should reduce this weighted interference relative to isotropic and suitably matched controls. That comparison, rather than the elementary inequality $N > d$, supplies a falsifiable geometric claim.

The chapter therefore advances two independent hypotheses. The first is that learned feature geometry can be evaluated against explicit random-code and coactivation-aware interference baselines. The second concerns the internal organization of polysemy: for some canonically polysemous tokens, context may be represented by symmetric operators that share a stable token-associated spectral projector while differing on low-dimensional sense subspaces in its orthogonal complement. These operators may fail to commute, and their

conditional projections define contextual embeddings. Concentration geometry does not imply this operator structure; it earns scientific content only if it improves held-out prediction or compression over dimension- and parameter-matched unstructured models. Section VD states how both hypotheses can fail.

A serious caveat is anisotropy. The nominal width of an LLM layer may exceed the dimension of the carrier actually occupied by its states. To avoid a third dimensional symbol, d throughout this chapter means the *usable exact dimension*: first restrict or whiten the representation to the carrier being tested, then let d be the size of an orthonormal basis of that carrier. If this usable d is small, the random-code crowding and readout baselines worsen. The method used to identify the carrier must therefore be declared empirically.

The reference-subspace and operator hypotheses have a narrower domain. They are meant for canonically polysemous tokens with established sense inventories, such as WordNet-, OntoNotes-, SemCor-, or WiC-style sense distinctions [3] with enough contextual instances for testing. Failure on a representative sample, against the controls in Sec. VD, would count against a privileged token reference subspace or shared-sector operator model even if the broader null calibration survives.

The proposal is adjacent to the superposition hypothesis in mechanistic interpretability [4], according to which neural networks represent more features than they have neurons by packing features into nearly orthogonal directions. The present chapter separates the geometric background from a possible semantic organization within it. The general superposition picture concerns an overcomplete dictionary of feature directions and sparse active sets. The present model adds the more specific possibility that contextual readout is organized around stable token-associated directions and context-dependent, potentially noncommuting projectors.

Section II develops the geometric null model: the deterministic coherence floor, an isotropic random-dictionary baseline, and the distinction between dictionary coexistence and simultaneous readout, culminating in the task-dependent criterion of Sec. IID. Section III introduces the reference-subspace hypothesis, its shared-sector projector and observable parameterization, and its relation to dynamic contextual embeddings. Section IV connects the construction to attention as soft context routing rather than literal basis reconstruction. Section V discusses superposition, sparse autoencoders, limitations, empirical tests, and the limited quantum analogy.

Global variables and local ranks. Throughout, $\mathbb{R}^d$ is the usable Euclidean carrier, and d is its exact integer dimension. The symbol N is the size of a prespecified semantic feature dictionary; it is a count of directions, not another dimension. A dictionary is called overcomplete when $N > d$, and its directions are called ε -quasi-orthogonal only if their measured overlaps satisfy the criterion below. The symbol k counts the directions active

in one context. The quantities ε and the later decision margin Δ are tolerances, while p and q_C are local ranks of particular reference and sense subspaces. None is a second global vector-space dimension.

The carrier has inner product $u^\top v$ and norm $\|v\| = \sqrt{v^\top v}$; $S^{d-1} = \{v \in \mathbb{R}^d : \|v\| = 1\}$ is the unit sphere; I is the identity matrix; $\|\cdot\|_{\text{op}}$ is the operator norm; $A \succeq 0$ means positive semidefinite; and $\mathbb{P}$ and $\mathbb{E}$ denote probability and expectation under the uniform measure on the relevant sphere unless stated otherwise. Universal positive constants are denoted c, c' , possibly changing from line to line.

II. A GEOMETRIC NULL MODEL FOR AN OVERCOMPLETE DICTIONARY

The results recalled in this section are standard facts about frames, spherical concentration, and correlation readout. No new packing theorem is claimed. Their role is to define a quantitative null model against which an empirically extracted feature dictionary can be compared. The quantities d, N , the metric, the extraction rule, and the readout must therefore be fixed before the comparison is made.

A. Overcomplete dictionaries and a hard coherence floor

Let $w_1, \dots, w_N \in \mathbb{R}^d$ be unit feature directions obtained by a prespecified procedure, and assume throughout this section that $N > d$. This is the domain under study, not a conclusion about semantic usefulness. The mutual coherence is

$$\mu(W) = \max_{i \neq j} |w_i^\top w_j|. \quad (3)$$

We call the dictionary ε -quasi-orthogonal when $\mu(W) \leq \varepsilon$. The term “pseudo-orthogonal” is avoided because it normally refers to an indefinite inner-product geometry.

Unlike the elementary observation that an overcomplete family cannot be mutually orthogonal, the Welch bound supplies a quantitative, deterministic coherence floor:

$$\mu(W)^2 \geq \frac{N-d}{d(N-1)} \quad (N > d) \quad (4)$$

for every unit-vector dictionary [5]. This is a genuine geometric constraint, but it is often too weak to predict the largest overlap of a particular learned dictionary. For that purpose we next introduce an isotropic random baseline.

B. Isotropic random-dictionary baseline

For independent uniform unit vectors $u, v \in S^{d-1}$,

$$\mathbb{E}[u^\top v] = 0, \quad \text{Var}(u^\top v) = \frac{1}{d}, \quad (5)$$

and spherical concentration gives, up to an inessential choice of constants,

$$\mathbb{P}\{|u^\top v| \geq \varepsilon\} \leq 2 \exp\left[-\frac{(d-1)\varepsilon^2}{2}\right] \quad (6)$$

for the small-overlap regime of interest [6, 7]. A union bound for N independently sampled directions yields

$$\mathbb{P}\{\mu(W) \geq \varepsilon\} \lesssim N^2 \exp(-cd\varepsilon^2). \quad (7)$$

Equation (7) is a sufficient high-probability statement for a random construction. It is not an upper bound on optimized spherical codes and is not a semantic impossibility theorem.

Ignoring polynomial prefactors in the spherical-cap tail, the typical largest overlap in the random ensemble satisfies

$$\mu_{\max}^{\text{null}}(d, N) \approx \sqrt{1 - N^{-4/d}} \simeq 2\sqrt{\frac{\log N}{d}}, \quad \log N \ll d. \quad (8)$$

The last expression is the small-overlap expansion used below. Learned directions are neither independent nor generally isotropic, so departures from Eq. (8) require matched controls rather than immediate interpretation as success or failure.

C. Dictionary coexistence versus simultaneous linear readout

Maximum coherence describes the dictionary, not the error of a specified decoder. Consider a context in which an active set S of size k contributes amplitudes x_j ,

$$h = \sum_{j \in S} x_j w_j. \quad (9)$$

For a target $i \in S$, the correlation readout is

$$w_i^\top h = x_i + \sum_{\substack{j \in S \\ j \neq i}} x_j w_i^\top w_j. \quad (10)$$

Under the isotropic or equivalent cancellation assumptions of Eq. (5), its conditional interference variance is

$$\text{Var}\left(\sum_{\substack{j \in S \\ j \neq i}} x_j w_i^\top w_j\right) \simeq \frac{1}{d} \sum_{\substack{j \in S \\ j \neq i}} x_j^2. \quad (11)$$

For order-one amplitudes, the fixed-target rms scale is therefore

$$\sigma_{\text{int}} \sim \sqrt{\frac{k-1}{d}} \simeq \sqrt{\frac{k}{d}}. \quad (12)$$

This is not a universal recovery limit. Without cancellation, coherence gives the worst-case estimate $|w_i^\top h - x_i| \leq \varepsilon \sum_{j \neq i} |x_j|$; checking many targets adds a multiple-comparison cost; and optimized sparse recovery invokes stronger conditions. For example, uniform recovery of a k -sparse coefficient vector typically requires a scaling such as $d \gtrsim Ck \log(N/k)$ under restricted-isometry assumptions [8–10]. The $\sqrt{k/d}$ law should therefore be tested only for the stated correlation readout or a demonstrably equivalent decoder.

D. A task-dependent operating criterion

The informal term “Goldilocks” refers here only to an already overcomplete dictionary whose interference remains tolerable for a specified task. It does not turn the definitional condition $N > d$ into a lower performance bound. Given a tolerated worst overlap ε and a fixed-target rms margin Δ , the isotropic null predicts the two separate scales

$$N_{\text{crowd}}(d, \varepsilon) \approx \exp\left(\frac{d\varepsilon^2}{4}\right), \quad k_{\text{rms}}(d, \Delta) \lesssim d\Delta^2. \quad (13)$$

The first is the order-of-magnitude crossover at which a random dictionary typically develops an ε -sized worst pair; the second concerns one specified linear readout. Neither is a universal upper bound on learned representations. In particular, a semantically relevant diagnostic should condition overlap on coactivation rather than penalize two features that are never required together.

For scale only, at $d = 4096$ the rms overlap of one isotropic pair is $1/\sqrt{d} = 1.6\%$. A dictionary with $N = 10^4$ contains almost fifty million pairs, and Eq. (8) gives a largest-overlap scale of about 9.5%. Setting $\varepsilon = 0.10$ places the random crossover near $N_{\text{crowd}} \approx 2.8 \times 10^4$. This is not the upper edge of a universal band beginning at 4096. At the same d , $k = 64$ has the separate fixed-target scale $\sqrt{k/d} = 0.125$.

Nothing in this null model is specifically semantic. Its scientific use is to support controlled comparisons: after d , N , the metric, activity, and amplitudes have been matched, does training redistribute overlap in a way that reduces contextual readout error? The next section introduces an independent, stronger hypothesis about how such context dependence might be organized for polysemous tokens.

III. TOKEN REFERENCE SUBSPACES AND SHARED-SECTOR CONTEXT OPERATORS

The previous section supplied classical geometric benchmarks under declared assumptions; it did not imply any particular organization of polysemy. We now ask a separate, more speculative question: does a trained model organize the contextual states of some polysemous tokens relative to a stable token-associated reference subspace? The hypothesis is scientifically useful only if that constraint generalizes to held-out data better than equally expressive unstructured alternatives.

The notation distinguishes mathematical types and roles. For a token t and a sense context C :

- $r_t \in \mathbb{R}^d$ is a unit candidate token reference vector;
- $\mathcal{R}_t \subset \mathbb{R}^d$ is the token reference subspace;
- P_t is the orthogonal projector onto $\mathcal{R}_t$;
- $S_{t,C} \subset \mathcal{R}_t^\perp$ is a low-dimensional sense subspace;
- $u_{t,C,j} \in S_{t,C}$ are vectors spanning that sense subspace;
- $h_\ell(t | x) \in \mathbb{R}^d$ is an observed contextual hidden state.

Thus r_t , $u_{t,C,j}$, and $h_\ell(t | x)$ are vectors, whereas $\mathcal{R}_t$ and $S_{t,C}$ are subspaces. The w_i used in Sec. II have a separate role: they are generic feature directions in the geometric null model and are not reused as sense labels.

A. The reference-subspace hypothesis

In the simplest version, let r_t be a candidate unit direction and set

$$\mathcal{R}_t = \text{span}\{r_t\}. \quad (14)$$

Empirically, r_t could be the normalized static input embedding, a normalized mean contextual embedding, or a learned token-specific probe direction. It is simply a hypothesized context-stable reference for token identity. It is not a mechanical or dynamical object. Any state can be decomposed relative to any direction, so the decomposition alone has no empirical content. The claim is that a token-associated $\mathcal{R}_t$ exposes sense structure better than matched control directions or subspaces.

For each context or sense label C , define a low-dimensional subspace

$$S_{t,C} = \text{span}\{u_{t,C,1}, \dots, u_{t,C,q_C}\}, \quad (15)$$

$$S_{t,C} \subset \mathcal{R}_t^\perp, \quad q_C \ll d - \dim(\mathcal{R}_t).$$

For the intended three-arm example, finance, river-edge, and seat/bench associations of “bank” correspond to different subspaces $S_{\text{bank},C}$ inside the common orthogonal carrier $\mathcal{R}_{\text{bank}}^\perp$. The verbal use in which an aircraft banks

or tilts supplies a fourth sense and would add a fourth context subspace; it is mentioned here but omitted from the drawing for clarity. A word sense is not a single vector: topic, syntax, selectional preferences, and world knowledge may require several spanning directions $u_{t,C,j}$.

The reference object may itself require more than one direction. In that case define

$$\mathcal{R}_t = \text{span}\{r_{t,1}, \dots, r_{t,p}\}, \quad p \geq 1, \quad (16)$$

and require

$$S_{t,C} \subset \mathcal{R}_t^\perp. \quad (17)$$

Equation (14) is the special case $p = 1$. The bare subspace construction specifies only a pattern of sharing. In quantum logic, different contexts can share one or more observables or spectral projections; finite-dimensional examples with two shared observables are known [11]. In the semantic model, $\mathcal{R}_t$ is a common token-associated reference subspace, while the sense-specific subspaces lie in its orthogonal complement. These incidence relations alone imply neither noncommutativity nor quantum probability. The next subsection makes the stronger, optional operator structure explicit.

The low-rank assumption is empirical. It predicts that dominant within-sense variation is low-dimensional after projection onto $\mathcal{R}_t^\perp$. This is compatible with evidence that contextualized embeddings can separate word senses, but it is stronger: it asks whether such separation is privileged relative to a token-associated reference subspace [12, 13].

B. Shared-sector context projectors and observables

The shared-subspace hypothesis can be parameterized by a constrained family of context projectors. Let $Q_t \in \mathbb{R}^{d \times p}$ have orthonormal columns spanning $\mathcal{R}_t$, so that $P_t = Q_t Q_t^\top$. For each context C , let $B_{t,C} \in \mathbb{R}^{d \times q_C}$ have orthonormal columns spanning $S_{t,C}$. Here Q_t and $B_{t,C}$ are basis matrices, not additional semantic directions.

The projector onto the token-plus-sense subspace is

$$\Pi_{t,C} = P_t + B_{t,C} B_{t,C}^\top. \quad (18)$$

All $\Pi_{t,C}$ have the common invariant sector $\mathcal{R}_t$ and differ on their sense sectors. In particular,

$$[\Pi_{t,C}, \Pi_{t,C'}] = [B_{t,C} B_{t,C}^\top, B_{t,C'} B_{t,C'}^\top] \quad (19)$$

need not vanish. Crucially, the common term P_t cancels from this commutator; sharing a sector does not cause noncommutativity. Moreover, two generic fitted projectors will often fail to commute, so a nonzero commutator alone is not evidence for a quantum-inspired account. It becomes discriminating only if its size predicts an independently specified order-sensitive contextual task or improves held-out prediction relative to matched projector models.

A symmetric context observable with the same basis-independent shared sector can be written

$$A_{t,C} = \lambda_t P_t + B_{t,C} \Lambda_{t,C} B_{t,C}^\top, \quad (20)$$

where the scalar λ_t is shared across contexts and the diagonal $\Lambda_{t,C}$ is context-specific. If $p > 1$, the eigenvalue λ_t is deliberately degenerate, so the observable identifies the shared subspace rather than an arbitrary basis within it. Additional spectral sectors may be included when an application requires a complete observable. The projector family is the empirical core of the proposal; the eigenvalues add scientific content only if they are prespecified or learned and improve a held-out observable outcome, rather than being assigned after the geometry is known.

The corresponding context-conditioned embedding is concrete:

$$h_{t,C}(x) = \Pi_{t,C} h(x) = P_t h(x) + B_{t,C} B_{t,C}^\top (I - P_t) h(x). \quad (21)$$

One may select a single C , or take a soft mixture of these embeddings as in Sec. IV. Equations (18)–(21) define a proposed constrained module or probe; they do not claim that current LLMs already implement it. The term “intertwined” is used in the contextuality-inspired incidence sense that several contexts share a spectral sector. It is not the standard operator-theoretic assertion that an intertwiner X satisfies $AX = XB$. Everything here uses ordinary real matrices on classical hardware.

C. Projection and gated subspace readout

Let $h(x) \in \mathbb{R}^d$ be a contextual state and retain the orthogonal projector P_t defined above. The two components are

$$\begin{aligned} h(x) &= h_{\text{ref}}(x) + h_\perp(x), \\ h_{\text{ref}}(x) &= P_t h(x), \\ h_\perp(x) &= (I - P_t) h(x) \in \mathcal{R}_t^\perp. \end{aligned} \quad (22)$$

The hypothesis is that most sense-disambiguating variation is present in $h_\perp(x)$, not that the decomposition itself is special. A basis-invariant score can be expressed directly with the orthonormal matrices already defined:

$$\begin{aligned} e_t(x) &= \|Q_t^\top h(x)\|^2 = \|P_t h(x)\|^2, \\ \beta_{t,C}(x) &= B_{t,C}^\top h_\perp(x), \\ f_{t,C}(x) &= \|\beta_{t,C}(x)\|^2 = \|B_{t,C} B_{t,C}^\top h_\perp(x)\|^2. \end{aligned} \quad (23)$$

Here e_t measures energy in the candidate reference subspace and $f_{t,C}$ measures projection energy in the candidate sense subspace. Both definitions work for any reference rank p and are invariant under changes of orthonormal basis within either subspace. An illustrative gated score is

$$s_{t,C}(x) = g_t(e_t(x)) f_{t,C}(x), \quad (24)$$

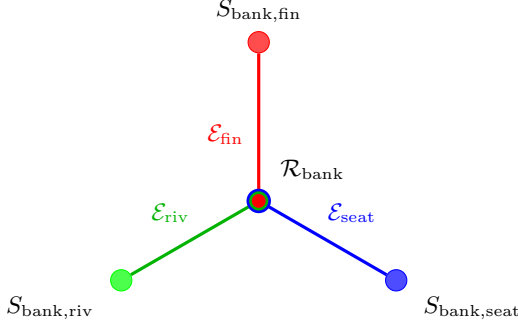


FIG. 1. Incidence schematic for the polysemous token “bank,” retaining the original star geometry. The center is the shared reference subspace $\mathcal{R}_{\text{bank}}$ and the outer nodes are the financial, river-edge, and seat/bench sense subspaces in $\mathcal{R}_{\text{bank}}^\perp$. Each colored spoke $\mathcal{E}_C$ labels a context incidence associated with $\Pi_{\text{bank},C}$; it is not a vector, map, or causal arrow. In a complete contextuality hypergraph, additional outcome sectors would complete each measurement context. The aircraft-banking sense could be added as a fourth spoke but is omitted for clarity.

where $g_t \geq 0$ is fixed or learned. The selected sense label is

$$C^*(x) = \arg \max_C s_{t,C}(x). \quad (25)$$

This score is an illustrative probe or router, not a derivation of how an LLM must disambiguate. At test time the true sense label is unavailable, so a practical implementation must learn the router from context without oracle access. The empirical content is the proposed separation of roles: the reference component tracks token identity or engagement, while the orthogonal component contains sense-disambiguating information. If no token-associated $\mathcal{R}_t$ exposes sense structure better than centered, token-mean, frequency-controlled, layer-matched, and random controls, the hypothesis fails.

Figure 1 illustrates these objects as an abstract hypergraph incidence structure.

D. Relation to dynamic contextual embeddings

Transformers already handle polysemy dynamically. The hidden state $h_\ell(t | x)$ of token t at layer ℓ depends on context x . The reference-subspace and shared-sector operator hypotheses are not alternative hardware architectures; they are proposed organizations and readouts of these observed trajectories.

Given a candidate reference subspace $\mathcal{R}_t$ and its projector P_t , write

$$\begin{aligned} h_\ell(t | x) &= P_t h_\ell(t | x) + z_\ell(t, x), \\ z_\ell(t, x) &= (I - P_t) h_\ell(t | x) \in \mathcal{R}_t^\perp. \end{aligned} \quad (26)$$

The decomposition is automatic; the hypothesis is not. It predicts that $P_t h_\ell(t | x)$ tracks token identity or engage-

ment, while $z_\ell(t, x)$ carries most sense-disambiguating information. Sense labels should be more separable in $\mathcal{R}_t^\perp$ than in the orthogonal complements of matched control subspaces. For each sense C , the projected states $z_\ell(t, x)$ should also concentrate near the low-dimensional subspace $S_{t,C} \subset \mathcal{R}_t^\perp$.

Thus the critical question is not whether contextual embeddings move; they do. It is whether their sense-relevant movement has a privileged organization relative to a token-associated reference subspace, and whether conditional application of $\Pi_{t,C}$ explains that movement better than matched unstructured projectors.

IV. ATTENTION AND LATENT CONTEXT SELECTION

A standard attention head computes

$$h' = \sum_j \alpha_j \nu_j, \quad \alpha_j = \frac{\exp(q_{\text{att}}^\top k_{\text{att},j} / \sqrt{d_{\text{head}}})}{\sum_{j'} \exp(q_{\text{att}}^\top k_{\text{att},j'} / \sqrt{d_{\text{head}}})}. \quad (27)$$

$q_{\text{att}}, k_{\text{att},j} \in \mathbb{R}^{d_{\text{head}}}$ are the query and key vectors of that head. The local head width d_{head} is an architectural quantity, not necessarily the selected carrier dimension d used in the geometric null model. Keys and values are produced by different learned maps, so value vectors are not generally coefficients of a dual basis for the key vectors. Attention is therefore not literal basis reconstruction.

The useful analogy is softer. Attention routes information: it amplifies some contextual directions and suppresses others. In the terminology of Sec. III, this resembles context-dependent subspace selection. A literal basis-reconstruction interpretation would require additional constraints, such as approximately Parseval keys and values implementing the corresponding dual reconstruction. We do not assume this.

A usable context selector cannot receive the true sense label at inference. Let a learned router $r_\theta(x, t)$ depend only on the available sequence and token state, and define

$$p_\theta(C | x, t) = \frac{\exp(r_{\theta,C}(x, t)/T)}{\sum_{C'} \exp(r_{\theta,C'}(x, t)/T)}. \quad (28)$$

The router may use the illustrative score in Eq. (24), but need not do so. Its weights provide a soft shared-sector projection:

$$\begin{aligned} \bar{h}_t(x) &= \sum_C p_\theta(C | x, t) h_{t,C}(x) \\ &= \sum_C p_\theta(C | x, t) \Pi_{t,C} h(x). \end{aligned} \quad (29)$$

Hard selection uses the largest router probability; soft selection uses Eq. (29). This is an ordinary mixture-of-projectors computation. Attention may supply its routing signal, but neither the shared sector nor the operator structure follows from standard attention. The

constrained model must therefore be compared with parameter-matched mixture-of-experts and unstructured low-rank alternatives.

V. DISCUSSION

The mathematical ingredients of the geometric analysis are standard coherence, concentration, and fixed-target readout estimates. Their role is not to turn the condition $N > d$ into a discovery, but to provide calibrated null predictions for an empirically extracted dictionary. The discriminating question is whether training systematically redistributes overlap: frequently coactive features should incur less interference, and held-out readout should improve relative to isotropic and activity-matched controls.

The shared-sector operator proposal is logically independent of that null model. It asserts that trajectories of some polysemous tokens admit a specific constrained organization. Neither a fitted common subspace nor a nonzero commutator is sufficient evidence. Evidence requires generalization to held-out contexts and an advantage over equally expressive unstructured projectors or mixture models.

A. Relation to superposition and sparse autoencoders

The superposition hypothesis holds that neural networks may represent more features than they have neurons by using nonorthogonal directions with sparse activity [4, 14]. The null model separates dictionary coexistence from simultaneous linear readout. In the isotropic, weak-dependence model, many directions can coexist while a fixed-target correlation decoder accumulates interference of scale $\sqrt{k/d}$. Sparse activity controls that interference for this decoder; it is not asserted to be universally necessary for structured dictionaries or nonlinear recovery.

Sparse autoencoders (SAEs) give a concrete empirical interface. They approximate layer activations as sparse expansions over learned decoder directions, making the w_i of Eq. (9) measurable objects rather than hypothetical features. Recent work has scaled SAEs, studied SAE feature geometry, and automated interpretation of large numbers of latents [15–18]. In the present framework, SAE decoder directions make the null calibration and coactivation-weighted diagnostic measurable, while sense-conditioned SAE activation clusters can test the stronger reference-subspace hypothesis.

B. A coactivation-weighted interference diagnostic

The active-set size k is a crude summary because it ignores which features appear together. For the fixed-

target correlation readout, define an ordered nonnegative pair weight

$$c_{ij} := \mathbb{P}(i, j \text{ active}) \mathbb{E}[x_j^2 \mid i, j \text{ active}] \omega_i, \quad (30)$$

where ω_i records the task importance of accurately reading feature i . Under the same zero-mean or weak-correlation assumptions used for the RMS calculation, a natural held-out diagnostic is

$$E_{\text{int}} = \sum_{i \neq j} c_{ij} (w_i^\top w_j)^2. \quad (31)$$

Unlike maximum coherence, Eq. (31) does not penalize a large overlap merely because it exists. If training or selection penalizes readout error, it predicts pressure toward smaller overlaps specifically for important features that coactivate. Mutually exclusive features can share geometric resources at much lower cost. This conditional redistribution, assessed against controls matched for marginal activity, amplitudes, feature importance, and dictionary size, is the paper’s substantive geometric prediction.

C. Limitations

The framework can fail in several ways.

First, the deterministic Welch bound constrains the coherence of a specified N -vector dictionary, but the crowding expression in Eq. (8) assumes independent isotropic directions. It is a typical null scale, not an optimal-code upper bound or a semantic impossibility theorem. The inequality $N > d$ is only the inclusion criterion for the overcomplete regime studied here; no lower usefulness boundary occurs at $N = d$.

Second, the nominal layer width need not equal the selected carrier rank d . Contextual embeddings can be anisotropic, and the occupied carrier can vary strongly by layer and training regime [19–22]. Whitening and related postprocessing can partially restore isotropic geometry [20, 21, 23]. At the same time, anisotropy should not be treated as an automatic or universal property of all Transformer representations [24]. The rule used to select or whiten a d -dimensional carrier must therefore be reported. Whitening also changes the metric in which overlaps are measured, and the covariance spectrum may contain information not summarized by one rank. We do not introduce a second dimension symbol, but d is protocol-dependent rather than an intrinsic semantic constant.

Third, N depends on the feature-extraction, normalization, and duplicate-removal rules. A claimed departure from the null may instead expose a poor dictionary, dependence among samples, finite-size effects, or a changed metric. These alternatives must be checked before attributing the departure to learned semantic organization.

Fourth, the reference-subspace pattern may simply be descriptively wrong. Trained models may represent polysemy without a preferred token-associated direction or

subspace. A low-rank fit found after inspecting test labels is not evidence; ranks, candidate references, and routers must be selected on training data and evaluated out of sample.

Fifth, the law $\sigma_{\text{int}} \sim \sqrt{k/d}$ concerns a fixed-target correlation readout with random-like directions and weakly correlated signs or amplitudes. It is neither a joint sparse-recovery theorem nor a universal equality. Structured dictionaries, nonlinear decoders, multiple comparisons over inactive features, and correlated errors can change the scaling.

Finally, the operator model has choices: the candidate reference subspace $\mathcal{R}_t$, the router, the local ranks, and any spectra in Eq. (20). The true sense label cannot be supplied to the router at test time. Eigenvalues are not identified by subspaces alone, and a fitted nonzero commutator is generic. These choices add evidence only through held-out prediction, compression, or prespecified order effects.

D. Empirical predictions

The key empirical task is to distinguish a null calibration from three substantive hypotheses.

Null calibration (not a semantic claim). For a dictionary independently established to be overcomplete, compare its coherence, overlap distribution, and fixed-target readout error with isotropic and activity-matched controls. Agreement with Eq. (8) is a baseline result, not evidence that overcompleteness is useful; departure from it must be diagnosed before being called learned structure.

Claim A: privileged token reference subspace. For a polysemous token t , there exists a privileged token-associated subspace $\mathcal{R}_t$ such that sense-relevant variation is better exposed in $\mathcal{R}_t^\perp$ than in the orthogonal complements of matched control subspaces of the same dimension.

Claim B: low-dimensional sense subspaces. Within $\mathcal{R}_t^\perp$, sense-conditioned states admit a cross-validated low-rank description that predicts or compresses held-out states better than shuffled-label, ordinary clustering, and matched-subspace controls. Related polysemous senses need not have minimal cross-sense coherence.

Claim C: shared-sector context operators. A non-oracular router and context projectors $\Pi_{t,C}$ sharing P_t improve held-out contextual prediction or compression relative to parameter-matched unstructured projectors and mixture-of-experts models. A commutator matters only if it predicts order effects defined independently of the fitted operators.

These claims can fail independently. The null calibration can hold while all three structural claims fail. Claim A can hold while sense information remains high-dimensional. Claim B can hold relative to a non-token-associated subspace, in which case Claim A fails. Claims A and B can hold even if the operator constraint in Claim

C adds no predictive value.

The following tests separate the possibilities.

1. *Null-model calibration.* Fix the carrier metric, spectral threshold or whitening rule, feature extractor, and duplicate criterion before evaluation. Report the selected rank d , dictionary size N , maximum coherence, the full overlap distribution, and active-set statistics. Compare with the Welch floor, isotropic dictionaries, and controls matched for marginal activity and coactivation. Treat Eq. (8) as a finite-sample null prediction, not a pass/fail capacity test.
2. *Reference-subspace test.* For a polysemous token, choose candidate reference subspaces generated by the normalized static input embedding, mean contextual embedding, learned token-specific directions, or a low-dimensional combination of these. Project layer states as in Eq. (26). Sense labels should be better separated in $\mathcal{R}_t^\perp$ than in orthogonal complements obtained after ordinary centering, token-mean subtraction, top-principal-component removal, and random, frequency-matched, or layer-matched control subspaces of the same rank. Compare held-out classification, compression, or incremental mutual information, with permutation over sense labels and bootstrap over token instances.
3. *Operational subspace test.* For each sense s , collect projected states $z_\ell(t, x) \in \mathcal{R}_t^\perp$ and estimate a low-rank subspace by PCA or probabilistic factor analysis. Select q_s without test-label leakage and use an orthonormal basis $B_{t,s}$. Evaluate held-out reconstruction and sense prediction against shuffled labels, ordinary supervised subspaces, and unsupervised clustering of equal rank. Report $\|B_{t,s}^\top B_{t,s'}\|_{\text{op}}$ descriptively, without assuming related senses must be nearly orthogonal.
4. *Gated-readout test.* Linear probes trained on $z_\ell(t, x)$ should perform comparably to probes trained on the full state $h_\ell(t | x)$, while probes restricted to the reference component $P_t h_\ell(t | x)$ should provide less incremental sense information after frequency, syntax, and sense-prior controls. Near-chance performance is not required because the reference component may legitimately retain correlated information.
5. *Shared-sector operator test.* Fit P_t , $B_{t,C}$, and the router on training contexts, then evaluate Eq. (29) without true sense labels on held-out instances. Compare with unconstrained low-rank projectors, ordinary contextual probes, parameter-matched mixture-of-experts models, and shuffled context labels. If an order claim is made, define the sequential linguistic or model intervention independently; then ask whether $\|[\Pi_{t,C}, \Pi_{t,C'}]\|_{\text{op}}$ predicts the measured asymmetry. Merely obtaining a nonzero commutator does not pass the test.

6. *Isotropic-null overlap test.* Independent isotropic directions have signed inner-product variance $1/d$. Compare the full pairwise distribution of token embeddings or SAE decoder directions with that null after the carrier and metric have been selected. Deviations may reflect dependence, anisotropy, extraction choices, or learned alignment; quasi-orthogonality alone does not predict variance $1/d$.
7. *Synthetic concurrency degradation.* At a fixed layer, inject controlled superpositions

$$h_k = h_0 + \lambda \sum_{j=1}^k x_j w_j, \quad (32)$$

using directions w_j obtained independently, for example from SAE decoder directions or frozen linear probes. As k increases, readout should degrade smoothly on the scale $\sqrt{k/d}$ for random-like directions under the specified correlation decoder. Include norm-preserving, random-direction, activation-frequency, and out-of-distribution controls; degradation by itself does not validate semantic superposition.

8. *Coactivation-dependent geometry.* At matched marginal activation frequency, feature similarity, SAE reconstruction error, amplitude, and task relevance, important frequently coactive features should contribute less to Eq. (31) than activity-matched controls. This tests learned redistribution of interference, independently of the reference-subspace hypothesis.

E. Scope of the quantum analogy

The analogy to quantum contextuality is structural and limited. Both settings can describe alternatives by observables with context-dependent eigenspaces. In the proposed construction, several contexts share the spectral projector P_t while differing on complementary sense sectors. We call this “intertwined” only in the explicitly declared contextuality-inspired incidence sense, not in the standard operator-theoretic sense. The shared P_t cancels from the commutator, so it is the relative geometry of the sense sectors that may produce order dependence. Figure 1 records the displayed incidence pattern; omitted complementary outcome sectors would be required to complete each measurement context in a full contextuality hypergraph. Equations (18)–(20) provide the corresponding operator parameterization. Vectors, subspaces, projectors, and observables are distinct mathematical types.

Quantum contextuality has the force of no-go theorems such as Kochen–Specker [25, 26]; no analogous theorem is claimed here for semantic embeddings. Real symmetric matrices can be noncommuting while still being stored

and evaluated on an ordinary classical computer, and generic fitted projectors are commonly noncommuting. Thus noncommutativity by itself supplies neither a semantic result, Born probabilities, nor physical quantumness.

Three levels of analysis should not be conflated.

1. *Classical high-dimensional representation.* A present-day Transformer carries real-valued hidden states through attention, nonlinear feed-forward maps, residual connections, and normalization. Concentration, quasi-orthogonality, feature superposition, dynamic embeddings, and finite readout are all available at this level. None requires quantum probability or quantum hardware.
2. *Quantum-like semantic or contextual formalism.* States, contexts, bases, interference terms, noncommuting questions, or measurement-like updates may be used as an abstract model of meaning even when the implementation remains classical. The relevant issue is not the vocabulary but whether this extra structure improves compression, explanation, or prediction relative to an ordinary contextual embedding model.
3. *Genuine quantum machine learning or quantum attention.* Here semantic states would be encoded in qubits or other physical quantum degrees of freedom, intermediate evolution would be unitary, phase would affect interference, and measurement would follow quantum probability. A standard Transformer does none of these things.

The random-dictionary and linear-readout baselines belong entirely to the first level. Shared-sector projection is also an ordinary classical computation. The proposal earns the stronger quantum-like interpretation only if its operator constraints or prespecified order effects improve prediction, compression, or explanation relative to equally expressive classical baselines. Even a positive result would support a useful formalism, not physical quantum computation; phase-sensitive dynamics and Born-rule measurement remain absent.

The dimension of $\mathcal{R}_t$ is also explicit. In Hilbert-space quantum logic, different contexts can share one observable, as in the familiar tripod-like case, or several observables; a four-dimensional example has two contexts with two common observables [11]. The semantic analogue is $\dim(\mathcal{R}_t) = p$: $p = 1$ is the simplest case, while $p > 1$ permits several token-associated reference directions. In every case, sense-specific information is hypothesized to occupy subspaces of $\mathcal{R}_t^\perp$.

VI. CONCLUSION

This chapter treats overcompleteness as an empirical starting condition, not as a theorem. After fixing a

carrier, metric, and feature-extraction rule, the dictionary under study has size $N > d$ in a carrier of rank d . The classical geometry then supplies calibrated baselines. The Welch bound is a genuine deterministic coherence floor for a specified N and d . By contrast, an isotropic random dictionary predicts a worst overlap of order $2\sqrt{\log N/d}$, and a random-like active set produces fixed-target linear readout interference of order $\sqrt{k/d}$. The associated scale $\exp(d\varepsilon^2/4)$ is a random-code crowding crossover, not a universal upper bound and not evidence that $N \leq d$ is semantically deficient.

The more consequential geometric hypothesis concerns where overlap is placed. Important features that frequently coactivate should contribute less coactivation-weighted interference than activity-, amplitude-, and importance-matched controls, whereas mutually exclusive features may share representational resources at lower cost. This turns standard geometry into a falsifiable claim about learned organization rather than a restatement of finite-dimensional linear algebra.

The second proposal is a constrained operator model of polysemy. Context projectors share a token-associated invariant sector and differ on low-dimensional sense subspaces in its orthogonal complement. Conditional projection yields a context-specific embedding, and a learned non-oracular router supplies hard or soft context selection. The common projector does not cause noncommutativity, and a nonzero fitted commutator has no evidential force by itself. The model succeeds only if the shared-sector constraint improves held-out prediction or compression over parameter-matched unstructured models, or if its commutators predict independently specified

order effects.

The null-model calibration and the shared-sector hypothesis can fail independently. A trained dictionary may redistribute interference without a privileged token reference subspace; conversely, a useful shared-sector model could exist even when global feature geometry is poorly described by an isotropic code. Keeping these claims separate preserves an informative outcome in either case.

The construction is quantum-inspired, not a claim that a Transformer is a quantum device. It uses real matrices on classical hardware and introduces no Born rule, unitary dynamics, or physical measurement. Quantum terminology is earned only insofar as the constrained operator structure provides additional, held-out explanatory or predictive value.

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